

LAB EXERCISES

Ex. No: 1

SIMPLE AND COMPOUND INTEREST

Aim

To write a C program to calculate the simple and compound interest.

Program

```

#include<stdio.h>
#include<math.h>
void main()
{
    int N;
    float P,R,SI,CI;
    clrscr();
    printf("Enter the principal amount\n");
    scanf("%f",&P);
    printf("\nEnter number of years\n");
    scanf("%d",&N);
    printf("\nEnter rate of interest\n");
    scanf("%f",&R);
    SI=(P*N*R)/100;
    CI=P*pow(1+R/100,N)-P;
    printf("\nSimple Interest = Rs.%6.2f",SI);
    printf("\nCompound Interest = %6.2f",CI);
    getch();
}

```

Sample Output

Enter the principal amount

20000

Enter number of years

5

Enter rate of interest

8

Simple Interest = Rs.8000.00

Compound Interest =Rs.9386.56

Result

Thus the above C program is executed and the output is obtained.

Ex. No: 2

AREA OF A CIRCLE AND A RECTANGLE

Aim

To write a C program to find the area of a circle and a rectangle (using preprocessing directory for defining pi value).

Program

```
#include<stdio.h>
#define PI 3.14
void main()
{
float r,l,b,cir_area,rect_area;
clrscr();
printf("Enter the value of radius r\n");
scanf("%f",&r);
cir_area=PI*r*r;
printf("Area of a circle is %6.2f",cir_area);
printf("\nEnter the value of length\n");
scanf("%f",&l);
printf("Enter the value of breadth\n");
scanf("%f",&b);
rect_area=l*b;
printf("Area of rectangle is %6.2f",rect_area);
getch();
}
```

Sample Output

```
Enter the value of radius r
10
Area of a circle is 314.00
Enter the value of length
20
Enter the value of breadth
30
Area of rectangle is 600.00
```

Result

Thus the above C program is executed and the output is obtained.

Ex. No: 3

L.3

LARGEST OF THREE NUMBERS

Aim

To write a C program to find the largest of three numbers.

Program

```
#include<stdio.h>
void main()
{
int a,b,c,large;
clrscr();
printf("Enter three numbers\n");
scanf("%d %d %d",&a,&b,&c);
if(a>b&&a>c)
large=a;
else if(b>c)
large=b;
else
large=c;
printf("\nThe largest among given three numbers = %d",large);
getch();
}
```

Sample Output

```
Enter three numbers
45 67 56
The largest among given three numbers = 67
```

Result

Thus the above C program is executed and the output is obtained.

L.4

Ex. No: 4

PRIME NUMBERS FROM 1 TO N

Aim

To write a C program to generate all prime numbers from 1 to N.

Program

```
#include<stdio.h>
void main()
{
    int n,i,j,r;
    clrscr();
    printf("\nGive the value of n\n");
    scanf("%d",&n);
    printf("\nThe prime numbers are\n");
    for(i=2;i<=n;i++)
    {
        for(j=2;j<=i;j++)
        {
            r=i%j;
            if(r==0)
                break;
        }
        if(j==i)
            printf("%d\t",i);
    }
    getch();
}
```

Sample Output

Give the value of n
30

The prime numbers are
2 3 5 7 11 13 17 19 23 29

Result

Thus the above C program is executed and the output is obtained.

L.5

Ex. No: 5

FACTORIAL USING RECURSION

Aim

To write a C program to find factorial of a given number using recursion.

Program

```
#include<stdio.h>
void main()
{
    long int fact(int);
    int n;
    long int f;
    clrscr();
    printf("\nEnter the value of n\n");
    scanf("%d",&n);
    f=fact(n);
    printf("\nThe factorial value of %d is %ld",n,f);
    getch();
}

long int fact(int n)
{
    long int f;
    if(n==0)
        return (1);
    else
        f=n*fact(n-1);
    return(f);
}
```

Sample Output

Enter the value of n
5

The factorial value of 5 is 120

Result

Thus the above C program is executed and the output is obtained.

Ex. No: 6

MATRIX ADDITION AND TRANSPOSE

Aim

To write a C program to demonstrate matrix addition and transpose operations.

Program

```
#include<stdio.h>
void main()
{
    int i,j,a[5][5],b[5][5],c[5][5],t1[5][5],t2[5][5],m,n;
    clrscr();
    printf("Enter the row value and column value\n");
    scanf("%d %d",&m,&n);
    printf("\nEnter the first matrix\n");
    for(i=0;i<m;i++)
    for(j=0;j<n;j++)
    scanf("%d",&a[i][j]);
    printf("\nEnter the second matrix\n");
    for(i=0;i<m;i++)
    for(j=0;j<n;j++)
    scanf("%d",&b[i][j]);
    //Matrix Addition
    for(i=0;i<m;i++)
    for(j=0;j<n;j++)
    c[i][j]=a[i][j]+b[i][j];
    printf("\nThe resultant matrix after addition is\n");
    for(i=0;i<m;i++)
    {
        for(j=0;j<n;j++)
        {
            printf("%d\t",c[i][j]);
        }
        printf("\n");
    }
    //Matrix Transpose
```

```
for(i=0;i<m;i++)
for(j=0;j<n;j++)
{
    t1[j][i]=a[i][j];
    t2[j][i]=b[i][j];
}
printf("Transpose of first matrix is\n");
for(i=0;i<n;i++)
{
    for(j=0;j<m;j++)
    {
        printf("%d\t",t1[i][j]);
    }
    printf("\n");
}
printf("Transpose of second matrix is\n");
for(i=0;i<n;i++)
{
    for(j=0;j<m;j++)
    {
        printf("%d\t",t2[i][j]);
    }
    printf("\n");
}
getch();
}
```

Sample Output

Enter the row value and column value

2 3

Enter the first matrix

1 2 3

4 5 6

Enter the second matrix

3 2 4

1 6 9

L.8

The resultant matrix after addition is

```
4  4  7
5  11 15
```

Transpose of first matrix is

```
1  4
2  5
3  6
```

Transpose of second matrix is

```
3  1
2  6
4  9
```

Result

Thus the above C program is executed and the output is obtained.

Ex. No: 7

LENGTH AND REVERSE OF A STRING USING POINTERS

Aim

To write a C program to find the length of a string and reverse a string using pointers.

Program

```
#include<stdio.h>
void main()
{
    char str[20];
    char *ptr,*str1,temp;
    int length=0,i;
    clrscr();
    printf("Enter a string\n");
    scanf("%s",str);
    //Finding length of a string
    ptr=str;
    while(*ptr!='\0')
    {
        length++;
        ptr++;
    }
```

L.9

```
}
printf("\nLength of a given string %s is %d",str,length);
//Reversing a string
str1=str;
ptr--;
for(i=0;i<(length-1)/2;i++)
{
    temp=*ptr;
    *ptr=*str1;
    *str1=temp;
    ptr--;
    str1++;
}
printf("\nThe reversed string is %s",str);
getch();
}
```

Sample Output

Enter a string

polytechnic

Length of a given string polytechnic is 11

The reversed string is cinhcetylop

Result

Thus the above C program is executed and the output is obtained.

Ex. No: 8

LINEAR SEARCH USING POINTER TO AN ARRAY

Aim

To write a C program for implementing linear search using pointer to an array.

Program

```
#include<stdio.h>
void main()
{
    int n,key,a[20],i,*ptr;
    clrscr();
```

L.10

```
printf("Enter the number of elements in the array\n");
scanf("%d",&n);
printf("Enter the elements of the array\n");
for(i=0;i<n;i++)
scanf("%d",&a[i]);
printf("\nEnter the element to search\n");
scanf("%d",&key);
for(ptr=a;ptr<a+n;ptr++)
{
    if(*ptr==key)
    {
        printf("\nThe given number %d is found in the array",key);
        printf("\nThe position of %d is %d",key,ptr-a+1);
        break;
    }
}
if(ptr-a==n)
printf("\nThe element %d is not found in the array",key);
getch();
}
```

Sample Output

```
Enter the number of elements in the array
5
Enter the elements of the array
10 20 30 40 50
Enter the element to search
40
The given number 40 is found in the array
The position of 40 is 4
```

Result

Thus the above C program is executed and the output is obtained.

L.11

Ex. No: 9

USING STRUCTURES

Aim

To write a C program to collect and print students details like name, marks, etc. and then to calculate total and average mark using structure.

Program

```
#include<stdio.h>
struct student
{
    int regno;
    char name[20];
    int mark[6];
    int total;
    float aver;
};
void main()
{
    struct student s[70];
    int n,i,j;
    clrscr();
    printf("\nEnter the number of students\n");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("\nEnter the register number, name and six marks one by one\n");
        scanf("%d %s",&s[i].regno,s[i].name);
        for(j=0;j<6;j++)
        {
            scanf("%d",&s[i].mark[j]);
        }
        s[i].total=0;
        for(j=0;j<6;j++)
        {
            s[i].total=s[i].total+s[i].mark[j];
        }
    }
}
```

L.12

```

}
s[i].aver=s[i].total/6.0;
printf("\n Total mark and average mark are \n");
printf("\n%d\t%s\t%d\t%.2f\n",s[i].regno,s[i].name,s[i].total,s[i].aver);
}
getch();
}

```

Sample Output

```

Enter the number of students
2
Enter the register number, name and six marks one by one
111 Kiran 50 60 70 80 90 90
Total mark and average mark are
111 Kiran 440 73.33
Enter the register number, name and six marks one by one
222 Kamal 45 55 65 75 85 95
Total mark and average mark are
222 Kamal 420 70.00

```

Result

Thus the above C program is executed and the output is obtained.

Ex. No: 10

USING FILE

Aim

To write a C program to count the number of characters, words and lines in a file.

Program

```

#include<stdio.h>
#include<stdlib.h>
void main()
{
FILE *f;
char ch;
int characters,words,lines;
clrscr();
f=fopen("sample.txt","r");

```

L.13

```

if(f==NULL)
{
printf("\nUnable to open file");
exit(EXIT_FAILURE);
}
characters=words=lines=0;
while((ch=fgetc(f))!=EOF)
{
characters++;
if(ch=='\n' || ch=='\0')
lines++;
if(ch==' ' || ch=='\t' || ch=='\n' || ch=='\0')
words++;
}
if(characters>0)
{
words++;
lines++;
}
printf("\nNumber of characters=%d",characters);
printf("\nNumber of words=%d",words);
printf("\nNumber of lines=%d",lines);
fclose(f);
getch();
}

```

Content of sample.txt file

C language is a structured language.
I love programming.
Programming with C is very interesting.

Sample Output

```

Number of characters=96
Number of words=15
Number of lines=3

```

Result

Thus the above C program is executed and the output is obtained.